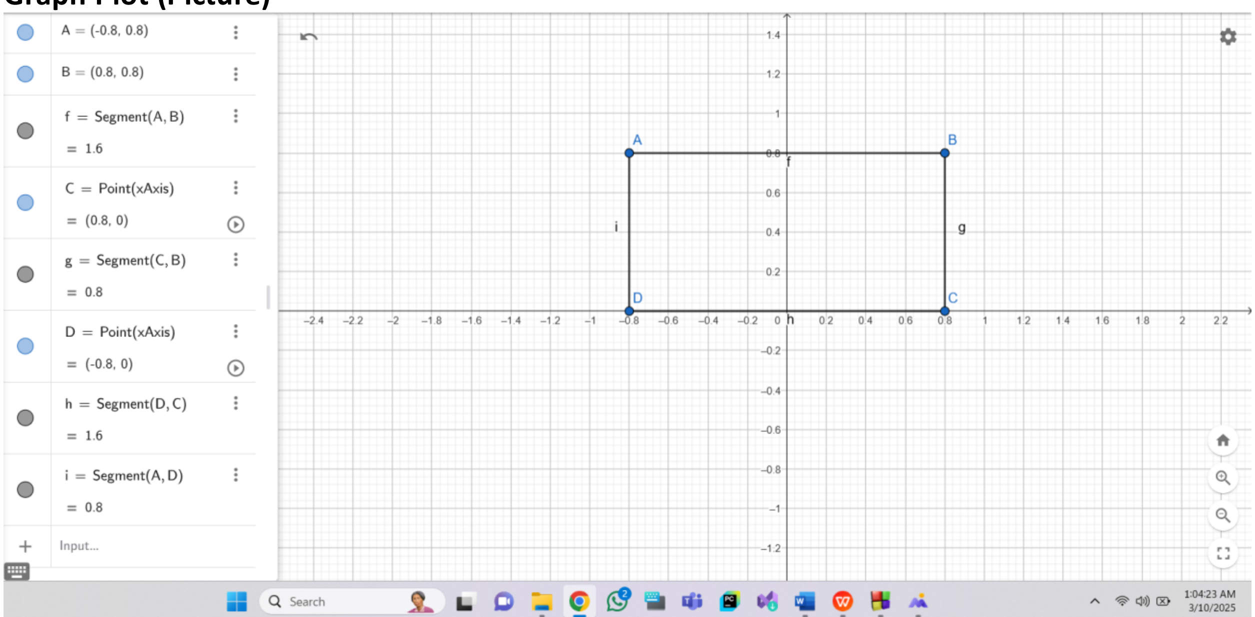
Lab Taks-1

Question-

Draw the object-

Graph Plot (Picture)-

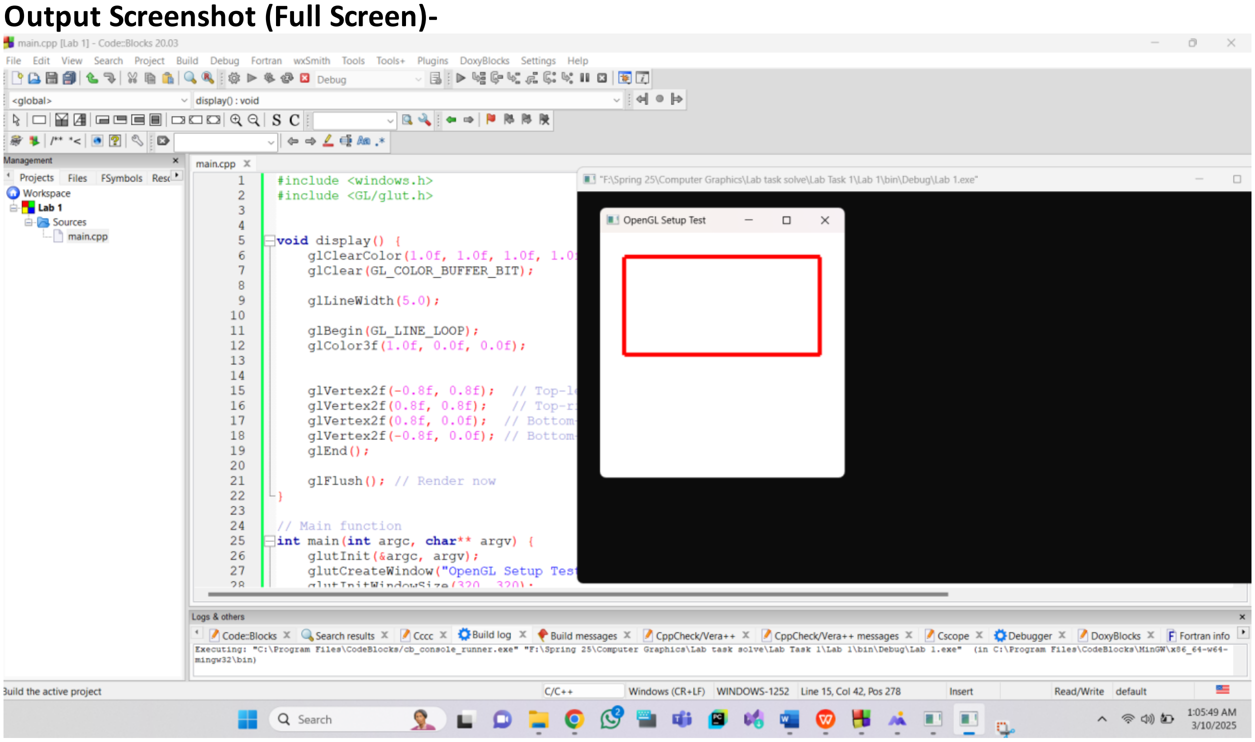


Code-

```
#include <windows.h>
#include <GL/glut.h>


void display() {
   glClearColor(1.0f, 1.0f, 1.0f, 1.0f);
   glClear(GL_COLOR_BUFFER_BIT);

   glLineWidth(5.0);

   glBegin(GL_LINE_LOOP);
```

```
    glColor3f(1.0f, 0.0f, 0.0f);


    glVertex2f(-0.8f, 0.8f);
    glVertex2f(0.8f, 0.8f);
    glVertex2f(0.8f, 0.0f);
    glVertex2f(-0.8f, 0.0f);
    glEnd();

    glFlush(); // Render now
}

// Main function
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutCreateWindow("OpenGL Setup Test");
    glutInitWindowSize(320, 320);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```

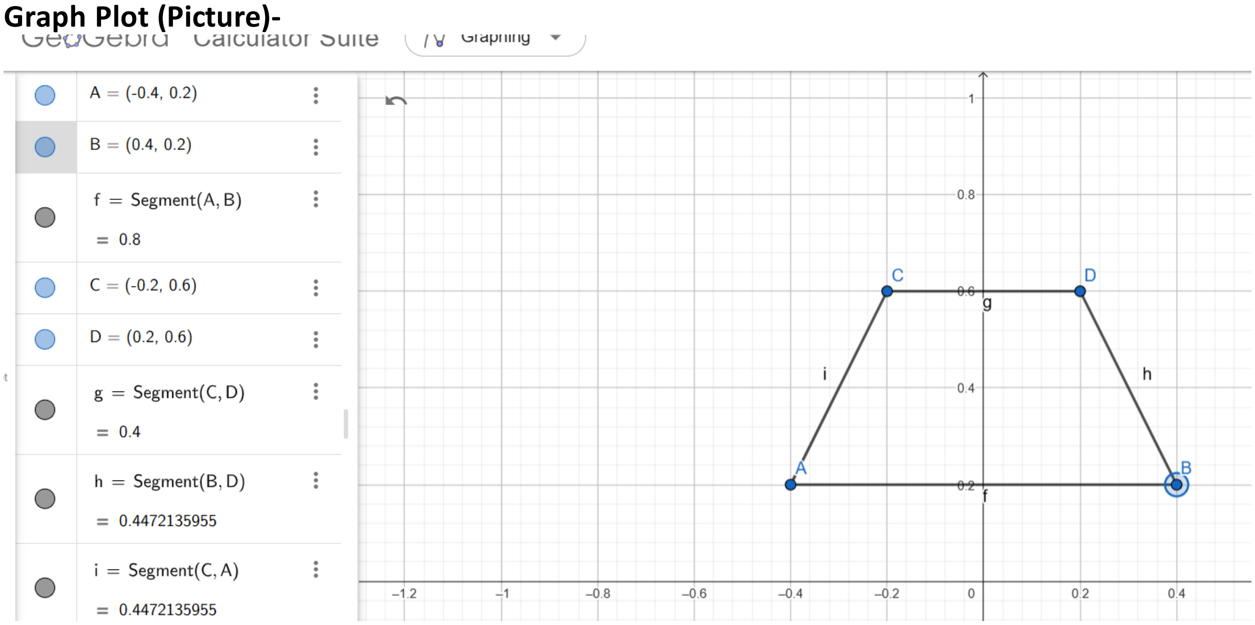
Output Screenshot (Full Screen)-



Question-

Draw the object-

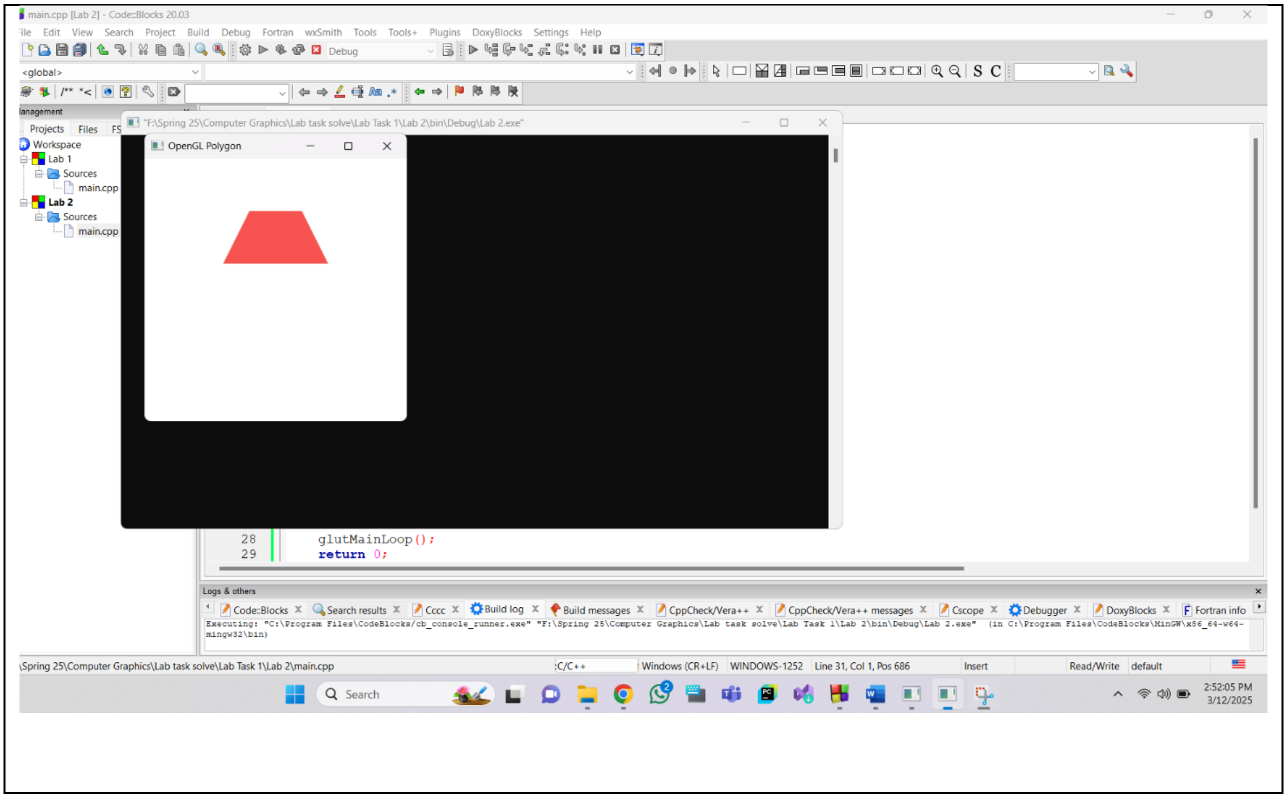
Graph Plot (Picture)-



Code-
```
#include <windows.h>
#include <GL/glut.h>

void display() {
    glClearColor(1.0f, 1.0f, 1.0f, 1.0f);
```

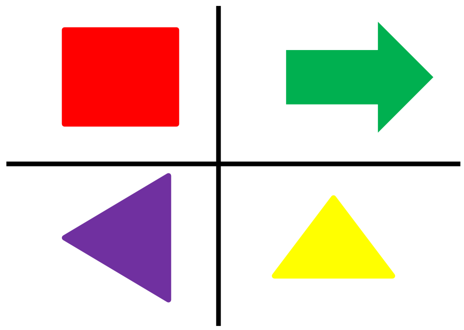
```
    glClear(GL_COLOR_BUFFER_BIT);

    glColor3ub(246, 82, 80); // Set polygon color

    glBegin(GL_POLYGON);
    glVertex2f(-0.4, 0.2);
    glVertex2f(0.4, 0.2);
    glVertex2f(0.2, 0.6);
    glVertex2f(-0.2, 0.6);
    glEnd();

    glutSwapBuffers(); // Use double buffering for smooth rendering
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(320, 320);
    glutCreateWindow("OpenGL Polygon");

    glutDisplayFunc(display);

    glutMainLoop();
    return 0;
}
```

Output Screenshot (Full Screen)-

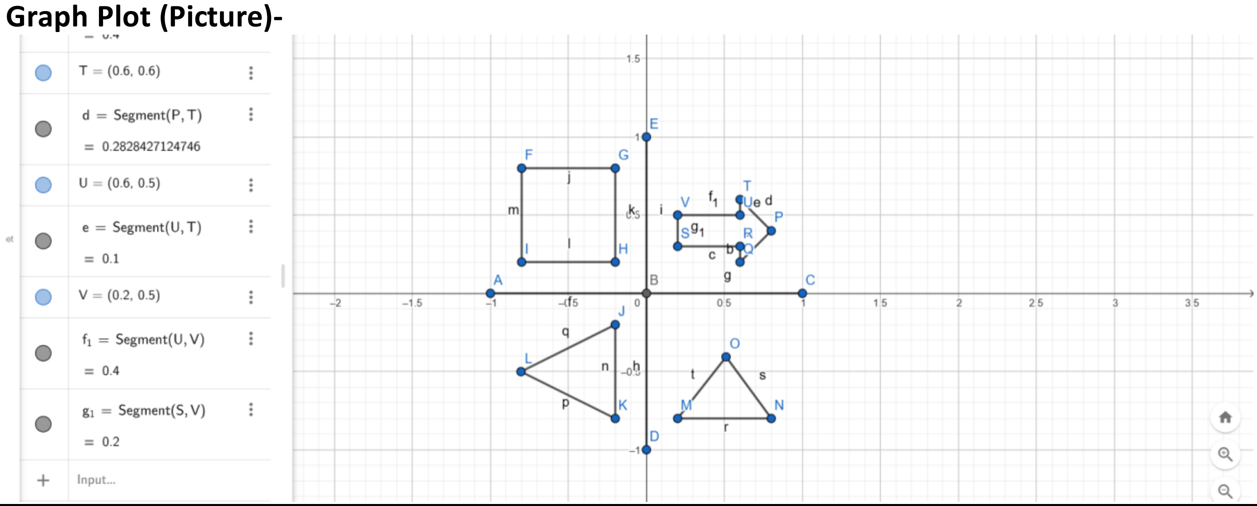

main.cpp [Lab 2] - Code::Blocks 20.03
"F:\Spring 25\Computer Graphics\Lab task solve\Lab Task 1\Lab 2\bin\Debug\Lab 2.exe"
OpenGL Polygon
Workspace
Lab 1
Sources
main.cpp
Lab 2
Sources
main.cpp
glutMainLoop();
return 0;
Logs & others
Executing: "C:\Program Files\CodeBlocks/cb_console_runner.exe" "F:\Spring 25\Computer Graphics\Lab task solve\Lab Task 1\Lab 2\bin\Debug\Lab 2.exe" (in C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\bin)
Line 31, Col 1, Pos 686
2:52:05 PM
3/12/2025

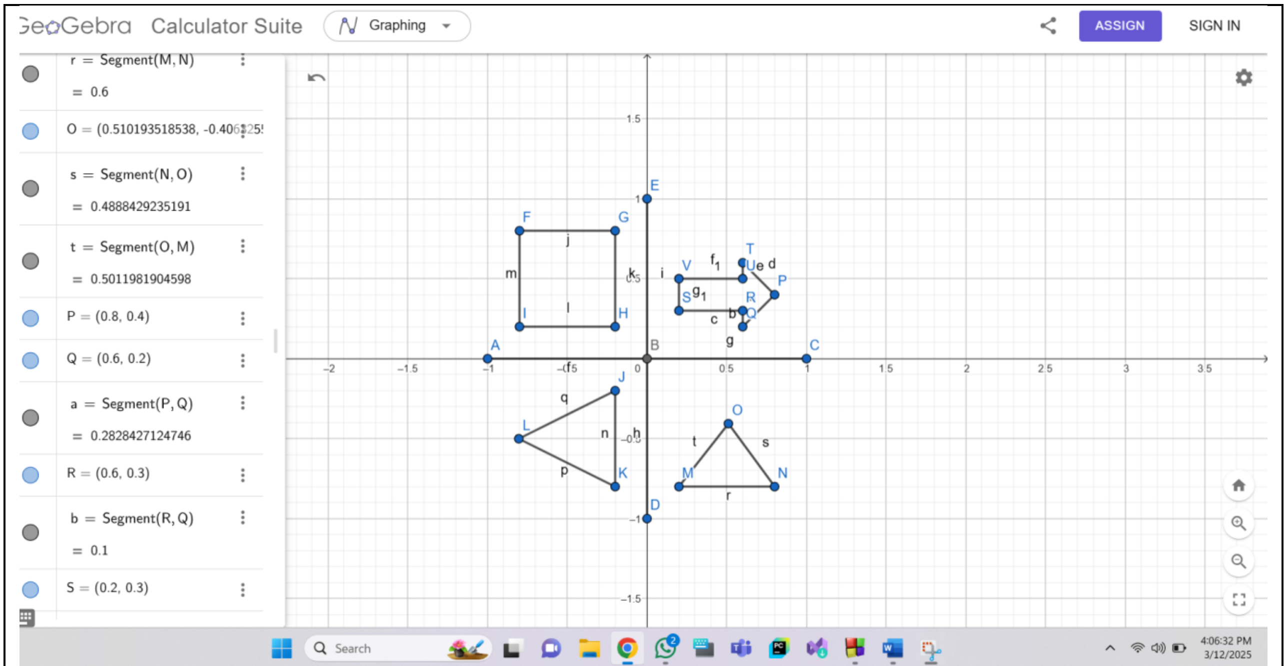
Question-

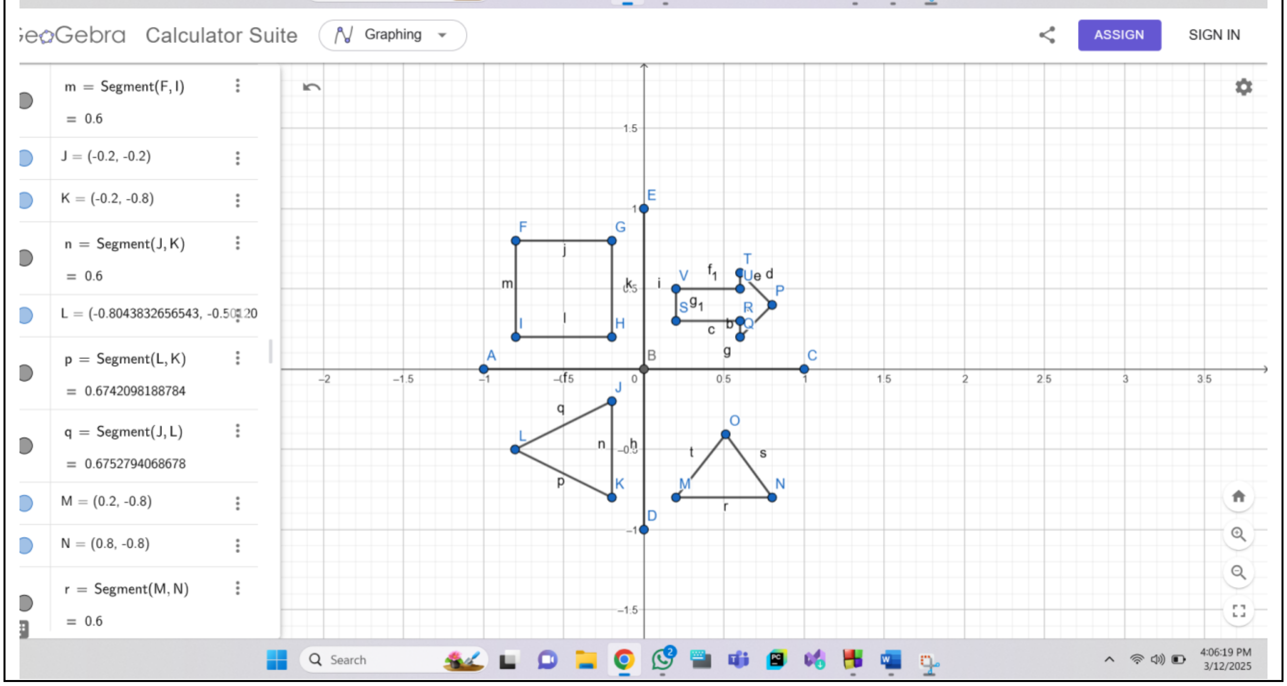
Draw the object-

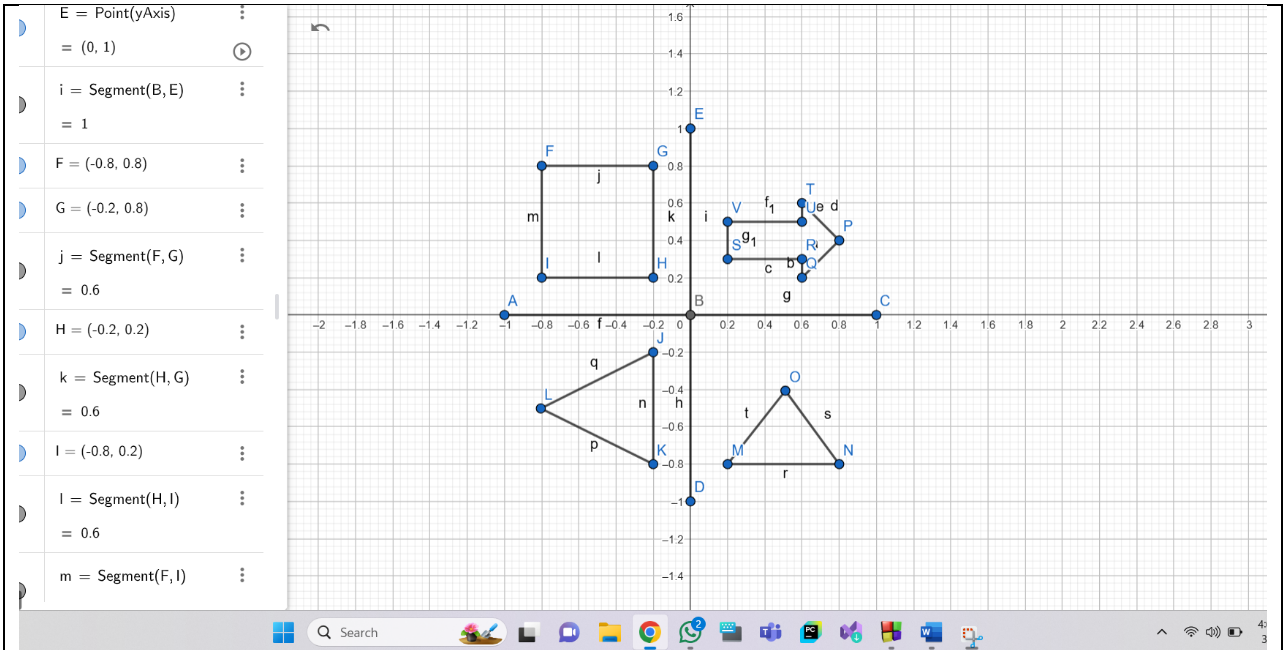


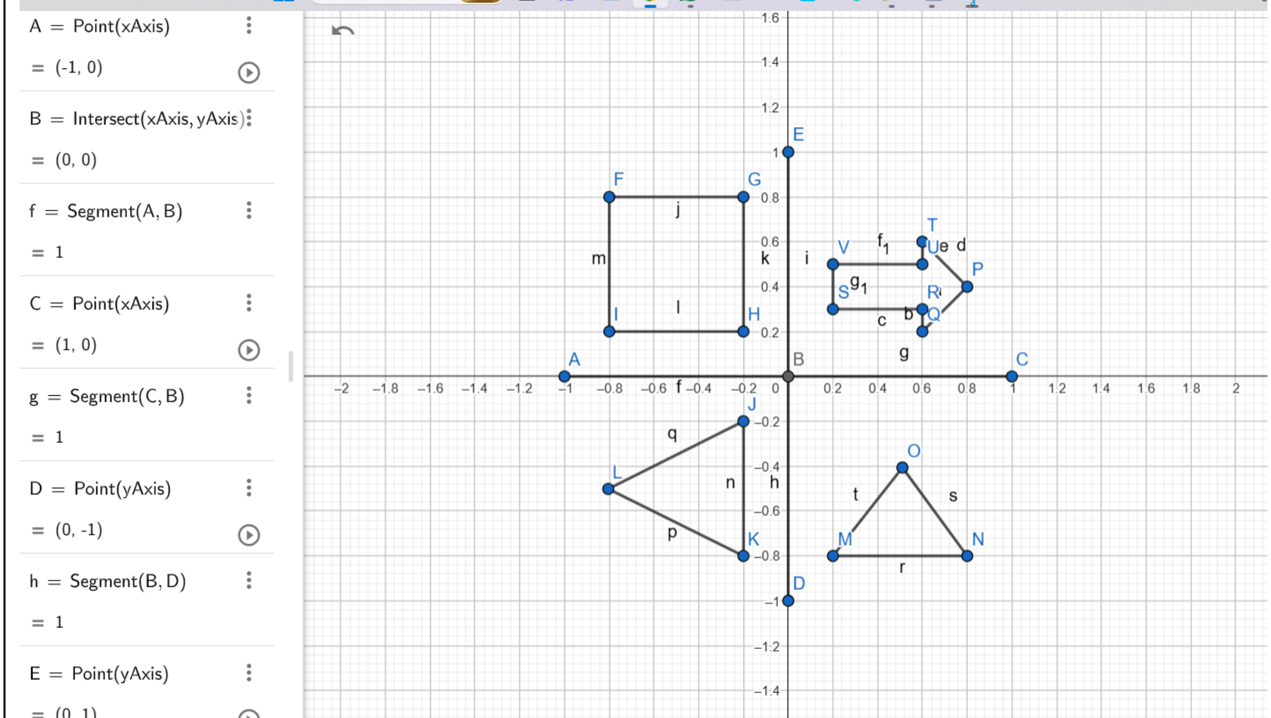
Graph Plot (Picture)-




GeoGebra Calculator Suite
Graphing
ASSIGN
SIGN IN
r = Segment(M, N)
= 0.6
O = (0.510193518538, -0.406
s = Segment(N, O)
= 0.4888429235191
t = Segment(O, M)
= 0.5011981904598
P = (0.8, 0.4)
Q = (0.6, 0.2)
a = Segment(P, Q)
= 0.2828427124746
R = (0.6, 0.3)
b = Segment(R, Q)
= 0.1
S = (0.2, 0.3)
4:06:32 PM
3/12/2025


GeoGebra Calculator Suite
Graphing
ASSIGN
SIGN IN
m = Segment(F, I)
= 0.6
J = (-0.2, -0.2)
K = (-0.2, -0.8)
n = Segment(J, K)
= 0.6
L = (-0.8043832656543, -0.5
p = Segment(L, K)
= 0.6742098188784
q = Segment(J, L)
= 0.6752794068678
M = (0.2, -0.8)
N = (0.8, -0.8)
r = Segment(M, N)
= 0.6
4:06:19 PM
3/12/2025



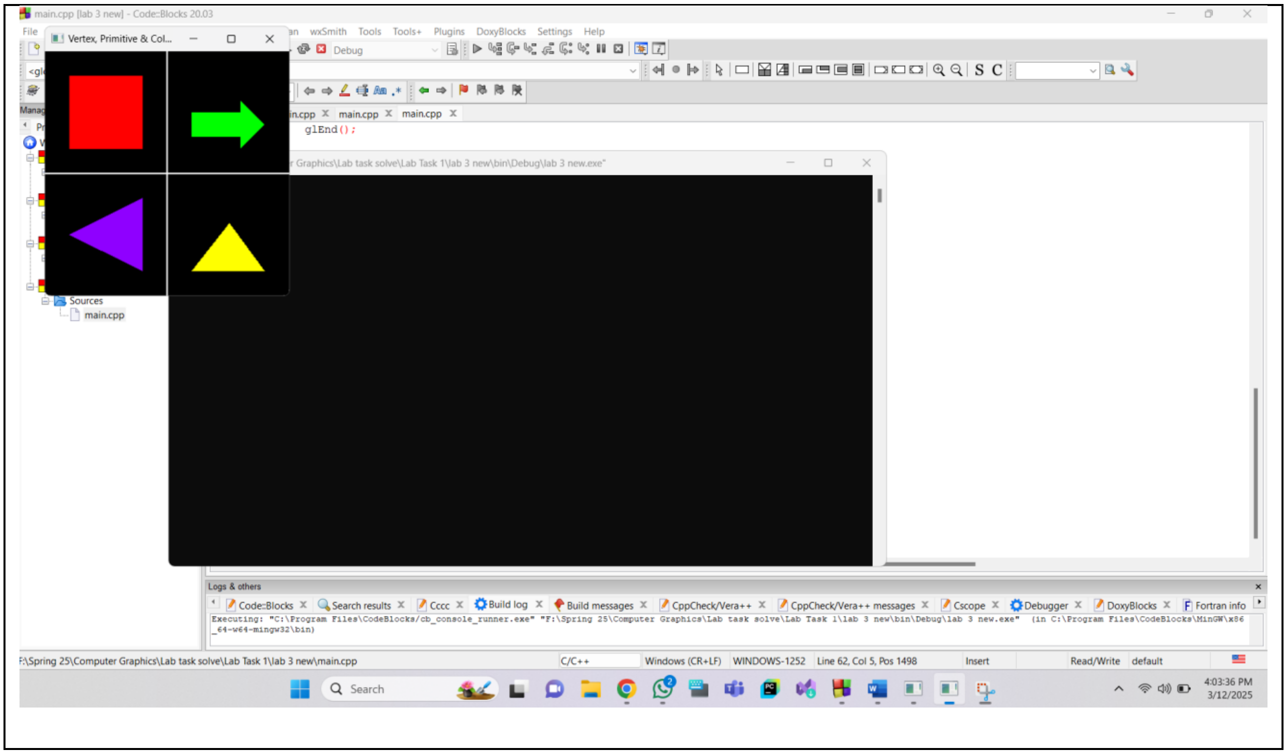


Code-
```
#include <windows.h>
#include <GL/glut.h>


void initGL() {
   glClearColor(0.0f, 0.0f, 0.0f, 1.0f); // Black background
}
```

```
void display() {
   glClear(GL_COLOR_BUFFER_BIT);

   // XY Axis
   glLineWidth(2.0);
   glBegin(GL_LINES);
   glColor3f(1.0f, 1.0f, 1.0f);

   // X-Axis
   glVertex2f(-1.0f, 0.0f);
   glVertex2f(1.0f, 0.0f);

   // Y-Axis
   glVertex2f(0.0f, -1.0f);
   glVertex2f(0.0f, 1.0f);
   glEnd();

   //  2nd Shape: Square

   glBegin(GL_QUADS);
   glColor3f(1.0f, 0.0f, 0.0f); // Red
   glVertex2f(-0.8f, 0.8f);  // Bottom-left
   glVertex2f(-0.2f, 0.8f);  // Bottom-right
   glVertex2f(-0.2f, 0.2f);  // Top-right
   glVertex2f(-0.8f, 0.2f);  // Top-left
   glEnd();

      //3rd Shape: TRIANGLES

   glBegin(GL_TRIANGLES);
   glColor3f(0.56f, 0.0f, 1.0f); // Violet
        glVertex2f(-0.2f, -0.2f);    // x, y
        glVertex2f(-0.804f, -0.501f);
        glVertex2f(-0.2f, -0.8f);

        glEnd();


        //  4th Shape: TRIANGLES

        glBegin(GL_TRIANGLES);
   glColor3f(1.0f, 1.0f, 0.0f); // Yellow
        glVertex2f(0.8f, -0.8f);    // x, y
```

```
        glVertex2f(0.2f, -0.8f);
        glVertex2f(0.510f, -0.406f);

        glEnd();


        //  1st Shape: Arrow
        glBegin(GL_POLYGON);
   glColor3f(0.0f, 1.0f, 0.0f); // Green
   glVertex2f(0.8f, 0.4f);  //P
   glVertex2f(0.6f, 0.2f); //Q
   glVertex2f(0.6f, 0.3f); //R
   glVertex2f(0.2f, 0.3f); //S
   glVertex2f(0.2f, 0.5f); //V
   glVertex2f(0.6f, 0.5f); //U
   glVertex2f(0.6f, 0.6f); //T

   glEnd();




   glFlush();  // Render now
}


int main(int argc, char** argv) {
   glutInit(&argc, argv);
   glutCreateWindow("Vertex, Primitive & Color");
   glutInitWindowSize(320, 320);
   glutDisplayFunc(display);
   initGL();
   glutMainLoop();
   return 0;
}
```

Output Screenshot (Full Screen)-



main.cpp [lab 3 new] - Code::Blocks 20.03
Vertex, Primitive & Col...
glEnd();
Sources
main.cpp
Logs & others
Executing: "C:\Program Files\CodeBlocks/cb_console_runner.exe" "F:\Spring 25\Computer Graphics\Lab task solve\Lab Task 1\lab 3 new\bin\Debug\lab 3 new.exe" (in C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\bin)
F:\Spring 25\Computer Graphics\Lab task solve\Lab Task 1\lab 3 new\main.cpp
Line 62, Col 5, Pos 1498
4:03:36 PM
3/12/2025